

Translating Compound Statements: Symbolic English

Logic & Reasoning Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Translate symbolic compound statements (using $\wedge$, $\vee$, $\neg$, $\rightarrow$, $\leftrightarrow$) into English sentences
- Translate English compound statements into symbolic form using correct logical operators
- Apply negation distribution and identify dominant operations in complex compound statements

Problems

1. Let P = 'It is raining' and Q = 'It is cold.' Translate the following symbolic statement into English.

2. Let P = 'She is tired' and Q = 'She is hungry.' Translate the following symbolic statement into English.

3. Let P = 'The sun is shining.' Translate the following symbolic statement into English.

$$\sim P$$

4. Let P = 'He is wealthy' and Q = 'He is happy.' Translate the following English statement into symbolic form: 'He is not wealthy and he is not happy.'

5. Let P = 'She is wealthy' and Q = 'She is happy.' Translate the following symbolic statement into English.

6. Let P = 'She is wealthy' and Q = 'She is happy.' Distribute the negation over the entire parenthetical expression and translate the result into English.

7. Let P = 'She is wealthy' and Q = 'She is happy.' Distribute the negation and translate the result into English.



8. Let P = 'A student misses a lecture', Q = 'A student studies', and R = 'A student fails.' Translate the following symbolic compound statement into English.

9. Let P = 'A student misses a lecture', Q = 'A student studies', and R = 'A student fails.' Translate the following English statement into symbolic form: 'A student studies, and if a student does not miss a lecture then a student does not fail.'

10. Let P = 'A student misses a lecture', Q = 'A student studies', and R = 'A student fails.' Translate the following complex symbolic compound statement into English, identifying the dominant operation first.



Translating Compound Statements: Symbolic English — Answer Key

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Answer Key

1. Answer: It is raining and it is cold.

- The symbol $\wedge$ represents the conjunction 'and'.
- P = 'It is raining', Q = 'It is cold'
- $P \wedge Q$ translates to: It is raining and it is cold.

2. Answer: She is tired or she is hungry.

- The symbol $\vee$ represents the disjunction 'or'.
- P = 'She is tired', Q = 'She is hungry'
- $P \vee Q$ translates to: She is tired or she is hungry.

3. Answer: The sun is not shining.

- The symbol $\sim$ (or $\neg$) represents negation, meaning 'not'.
- P = 'The sun is shining'
- $\sim P$ translates to: The sun is not shining.

4. Answer: $\sim P \wedge \sim Q$

- Identify each simple statement: P = 'He is wealthy', Q = 'He is happy'
- Negate P : $\sim P$ = 'He is not wealthy'
- Negate Q : $\sim Q$ = 'He is not happy'
- Join with 'and' ($\wedge$): $\sim P \wedge \sim Q$

5. Answer: She is not wealthy and she is happy.

- $\sim P$ negates only P : 'She is not wealthy'
- Q remains unchanged: 'She is happy'
- $\wedge$ connects them with 'and'
- Result: She is not wealthy and she is happy.

6. Answer: She is neither wealthy nor happy. (Symbolic: $\sim P \wedge \sim Q$)

- Distribute $\sim$ over the disjunction: $\sim(P \vee Q) = \sim P \wedge \sim Q$ (De Morgan's Law)
- $\sim P$ = 'She is not wealthy', $\sim Q$ = 'She is not happy'
- The conjunction of two negations can be phrased as 'neither...nor...'
- Result: She is neither wealthy nor happy.

7. Answer: She is not wealthy or she is not happy. (Symbolic: $\sim P \vee \sim Q$)

- Apply De Morgan's Law: $\sim(P \wedge Q) = \sim P \vee \sim Q$
- $\sim P$ = 'She is not wealthy', $\sim Q$ = 'She is not happy'
- $\vee$ connects them with 'or'



- Result: She is not wealthy or she is not happy.

8. Answer: If a student studies and does not miss a lecture, then the student does not fail.

- Identify the dominant operation: the conditional arrow ($\rightarrow$) separates antecedent and consequent.
- Antecedent: $Q \wedge \sim P$ = 'A student studies and does not miss a lecture'
- Consequent: $\sim R$ = 'A student does not fail'
- Combine using 'if...then...': If a student studies and does not miss a lecture, then the student does not fail.

9. Answer: $Q \wedge (\sim P \rightarrow \sim R)$

- Identify the dominant operation: 'and' ($\wedge$) connects two parts.
- First part: 'A student studies' = Q
- Second part: 'if a student does not miss a lecture then a student does not fail' = $\sim P \rightarrow \sim R$
- Combine: $Q \wedge (\sim P \rightarrow \sim R)$

10. Answer: A student misses a lecture or studies if and only if a student does not fail and does not miss a lecture.

- Identify the dominant operation: biconditional ($\leftrightarrow$) — 'if and only if'
- Left side: $P \vee Q$ = 'A student misses a lecture or a student studies'
- Right side: $\sim R \wedge \sim P$ = 'A student does not fail and does not miss a lecture'
- Combine using 'if and only if': A student misses a lecture or studies if and only if a student does not fail and does not miss a lecture.

